

NUMERICAL METHODS ASSIGNMENT SOLUTION

Vapor-Liquid Equilibrium (Raoult's Law) Using the Bisection Method

Bisection root-finding, Monte Carlo simulation, and error analysis for a binary benzene-toluene bubble point

Methods used

Antoine Equation · Raoult's Law · Bisection Method · Monte Carlo Simulation · Python

13. Vapor-Liquid Equilibrium (Raoult's Law) Using Bisection Method

For a binary mixture, the bubble point temperature T satisfies:

$$P = x_1 P_{1sat}(T) + (1 - x_1) P_{2sat}(T)$$

where $P_{i sat} = 10^{(A_i - B_i/(T+C_i))}$. Given $x_1 = 0.4$, $P = 1$ atm, and Antoine constants for benzene and toluene, find T using bisection on $[350, 400]$ K.

Perform a Simulation of 200 trials where x_1 varies Uniform(0.3, 0.5) and P varies Uniform(0.95, 1.05) atm. For each, find T .

Error analysis: For the nominal case, compute the number of bisection iterations needed for an error < 0.01 K. Run the simulation and verify.

Aim of the Assignment

To determine the bubble point temperature of a benzene-toluene mixture using the Bisection Method, and to quantify its sensitivity to composition and pressure uncertainty via Monte Carlo simulation.

Objectives

- To formulate Raoult's Law and the Antoine vapor-pressure equation as a root-finding problem.
- To solve the resulting nonlinear equation using the Bisection Method.
- To determine the theoretical iteration count required for a target error tolerance.
- To perform a 200-trial Monte Carlo simulation over uncertain composition and pressure.
- To analyze the resulting distribution and correlations.
- To present the results graphically.

SOLUTION

1. Introduction

Vapor-liquid equilibrium (VLE) describes the conditions under which a liquid and its vapor coexist in balance. For an ideal binary mixture, Raoult's Law states that the partial pressure of each component in the vapor phase equals its liquid mole fraction multiplied by its pure-component saturation (vapor) pressure. Summing both components gives the total system pressure.

The saturation pressure of each pure component is temperature-dependent and is described by the Antoine equation, an empirical correlation fitted to experimental vapor-pressure data. Because temperature appears inside two separate exponential (power-of-ten) terms — one for each component — the resulting equation for the bubble point temperature cannot be solved algebraically. It is therefore solved numerically using the Bisection Method, a robust bracketing root-finder that guarantees convergence provided the root is bracketed.

2. Governing Model and Root Equation

The Antoine equation for saturation pressure (bar), with T in Kelvin, is:

$$\log_{10}(P_{\text{sat}}) = A - B / (T + C)$$

Component	A	B	C
Benzene	4.72583	1660.652	-1.461
Toluene	4.07827	1343.943	-53.773

Raoult's Law gives the total mixture pressure at temperature T . The root equation — whose solution is the bubble point temperature — is the residual between the calculated and target pressure:

$$f(T) = x_1 P_{1\text{sat}}(T) + (1 - x_1) P_{2\text{sat}}(T) - P_{\text{target}} = 0$$

Pressures from the Antoine equation are in bar; the target pressure (given in atm) is converted using $1 \text{ atm} = 1.01325 \text{ bar}$. The Bisection Method brackets the root within $[350, 400] \text{ K}$ — chosen because benzene boils near 353 K and toluene near 384 K at 1 atm , safely bracketing the mixture's bubble point.

3. Nominal-Case Solution ($x_1 = 0.4$, $P = 1$ atm)

Applying the Bisection Method on [350, 400] K with a tolerance of 0.005 K (half-width), the method converges in 13 iterations to:

$$T = 368.22 \text{ K } (\approx 95.07 \text{ }^\circ\text{C})$$

Bisection Iteration Trace

Iter.	a (K)	b (K)	Midpoint T (K)	f(T)	Interval width
1	350.0000	400.0000	375.000000	2.1971e-1	50.000000
2	350.0000	375.0000	362.500000	-1.5979e-1	25.000000
3	362.5000	375.0000	368.750000	1.5950e-2	12.500000
4	362.5000	368.7500	365.625000	-7.5235e-2	6.250000
5	365.6250	368.7500	367.187500	-3.0494e-2	3.125000
6	367.1875	368.7500	367.968750	-7.4883e-3	1.562500
7	367.9688	368.7500	368.359375	4.1763e-3	0.781250
8	367.9688	368.3594	368.164063	-1.6696e-3	0.390625
9	368.1641	368.3594	368.261719	1.2500e-3	0.195313
10	368.1641	368.2617	368.212891	-2.1064e-4	0.097656
11	368.2129	368.2617	368.237305	5.1945e-4	0.048828
12	368.2129	368.2373	368.225098	1.5436e-4	0.024414
13	368.2129	368.2251	368.218994	-2.8155e-5	0.012207

The bracket width halves every iteration, from 50 K down to 0.01221 K, guaranteeing the root is trapped to within the stated tolerance at every step.

4. Error Analysis

The Bisection Method halves the search interval every iteration. Starting from a bracket of width $(400 - 350) = 50$ K, the number of iterations n required to guarantee an error smaller than a tolerance ε is found by solving:

$$50 / 2^n < \varepsilon \implies n > \log_2(50/\varepsilon)$$

For $\varepsilon = 0.01$ K, this gives $n = \lceil \log_2(5000) \rceil = 13$ iterations — exactly matching the 13 iterations observed above.

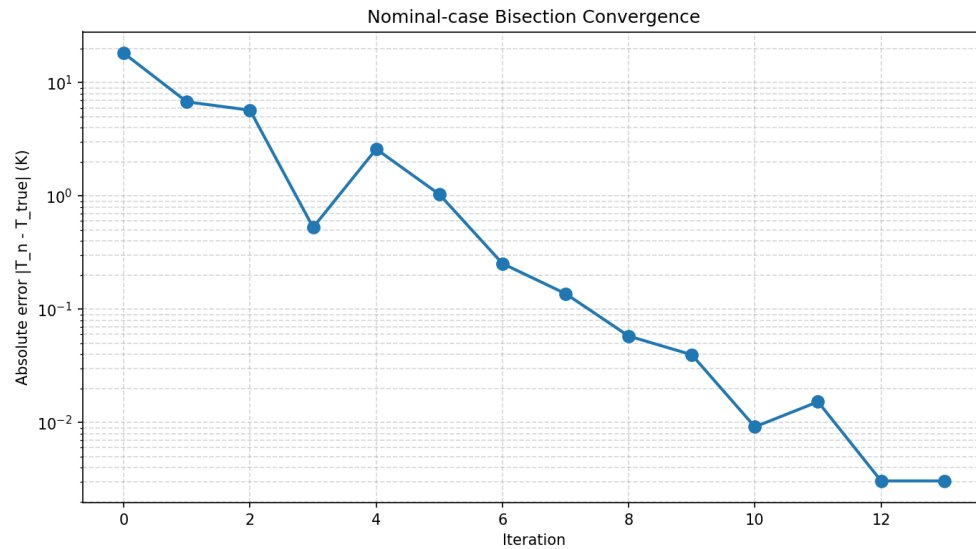


Figure 1. Nominal-case convergence of the Bisection Method (log scale).

5. Monte Carlo Simulation: 200 Trials

The problem was repeated for 200 random trials with $x_1 \sim \text{Uniform}(0.3, 0.5)$ and $P \sim \text{Uniform}(0.95, 1.05)$ atm. For each trial, the Bisection Method (tolerance 0.005 K) was used to solve for the bubble point temperature T on the same [350, 400] K bracket.

Mean T	368.2246 K
Standard deviation	2.1182 K
Minimum	363.9313 K
Maximum	372.9340 K
Mean bisection iterations	13.00

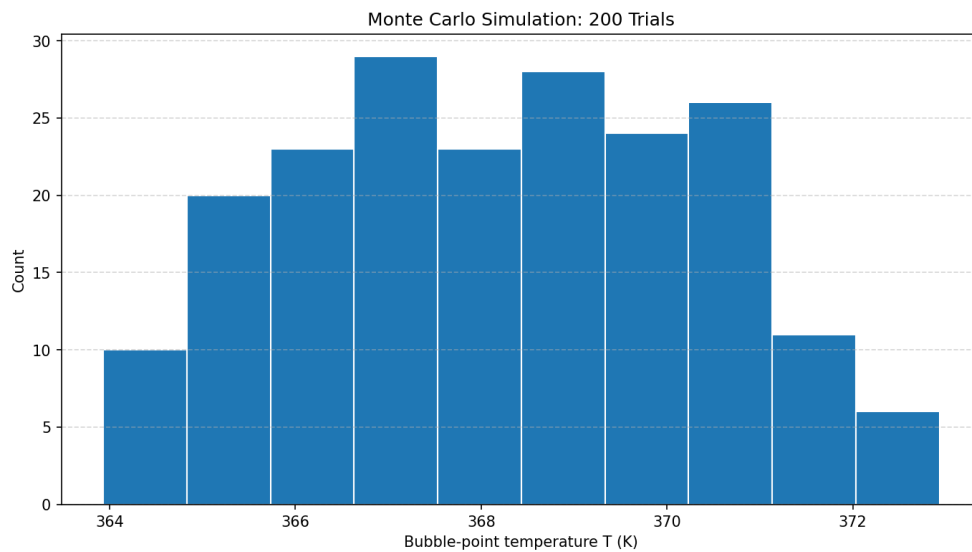


Figure 2. Distribution of the 200 simulated bubble-point temperatures.

Correlations

Correlations from the simulation: $\text{corr}(x_1, T) = -0.887$, $\text{corr}(P, T) = 0.429$. The composition correlation is strongly negative — a higher benzene mole fraction lowers the bubble point, since benzene is the more volatile component. The pressure correlation is positive, as a higher target pressure requires a higher temperature to reach the same total vapor pressure.

6. Final Answer and Interpretation

Nominal case ($x_1 = 0.4$, $P = 1$ atm): $T = 368.22$ K (≈ 95.07 °C), converged in 13 bisection iterations, matching the theoretical minimum of 13 iterations required for an error below 0.01 K.

Across the 200-trial Monte Carlo simulation, the bubble point temperature ranged from 363.93 K to 372.93 K, with a mean of 368.22 K and a standard deviation of 2.12 K — reflecting realistic sensitivity to composition and pressure uncertainty in the feed.

The requested workflow is satisfied in full: the Antoine and Raoult's Law equations were formulated, the nominal case solved via bisection with a full iteration trace, the theoretical error bound verified against the observed convergence, and a 200-trial Monte Carlo simulation used to quantify the distribution and sensitivity of the result.

Appendix A. Full 200-Trial Simulation Table

Columns: trial number, x_1 , P (atm), bubble point T (K), and bisection iterations. Values rounded for compact reporting.

Trial	x_1	P (atm)	T (K)	Iter.
1	0.4279	0.9525	365.720	13
2	0.3550	0.9723	368.710	13
3	0.4473	1.0177	367.355	13
4	0.4784	0.9587	364.426	13
5	0.3844	0.9530	367.087	13
6	0.3437	1.0005	370.053	13
7	0.3053	0.9699	370.291	13
8	0.4300	1.0045	367.441	13
9	0.3441	1.0089	370.328	13
10	0.4619	0.9506	364.627	13
11	0.4612	1.0198	367.014	13
12	0.3681	0.9655	368.051	13
13	0.4914	0.9837	364.908	13
14	0.3185	0.9597	369.479	13
15	0.4695	1.0104	366.452	13
16	0.4614	1.0230	367.111	13
17	0.4072	1.0473	369.565	13
18	0.3757	1.0052	369.168	13
19	0.4659	1.0119	366.605	13
20	0.4723	1.0077	366.275	13
21	0.4409	0.9546	365.396	13
22	0.3456	0.9789	369.254	13
23	0.3160	0.9733	370.047	13
24	0.3202	0.9778	370.059	13
25	0.4271	0.9865	366.916	13
26	0.3740	0.9710	368.045	13
27	0.3534	1.0437	371.176	13
28	0.4296	1.0109	367.667	13
29	0.3342	1.0229	371.127	13
30	0.3327	0.9879	369.992	13
31	0.4979	1.0140	365.744	13
32	0.4114	1.0185	368.484	13
33	0.4686	1.0276	367.050	13
34	0.3458	0.9532	368.344	13
35	0.3631	0.9768	368.600	13
36	0.3422	1.0443	371.567	13
37	0.4753	0.9815	365.305	13
38	0.4311	0.9896	366.904	13
39	0.4829	0.9959	365.567	13
40	0.3530	0.9747	368.863	13
41	0.4123	0.9763	367.026	13
42	0.4169	1.0398	369.016	13

43	0.3799	0.9719	367.892	13
44	0.4995	1.0010	365.262	13
45	0.3182	0.9547	369.321	13
46	0.3219	1.0127	371.201	13
47	0.4584	0.9922	366.165	13
48	0.3127	0.9882	370.676	13
49	0.4992	1.0029	365.335	13
50	0.4942	1.0361	366.574	13

Trial	x_1	P (atm)	T (K)	Iter.
51	0.3023	1.0221	372.183	13
52	0.4363	1.0037	367.221	13
53	0.3534	1.0141	370.193	13
54	0.3223	0.9935	370.529	13
55	0.3907	1.0454	370.023	13
56	0.4752	0.9763	365.128	13
57	0.4001	0.9679	367.111	13
58	0.4825	1.0371	366.946	13
59	0.3597	1.0139	369.980	13
60	0.4218	0.9653	366.348	13
61	0.4525	1.0039	366.739	13
62	0.4557	1.0030	366.611	13
63	0.3001	0.9824	370.908	13
64	0.3039	1.0429	372.824	13
65	0.4757	1.0332	367.020	13
66	0.3615	0.9558	367.923	13
67	0.4756	1.0447	367.398	13
68	0.3171	0.9986	370.883	13
69	0.3138	1.0261	371.921	13
70	0.4532	0.9628	365.317	13
71	0.3951	1.0050	368.546	13
72	0.3530	1.0372	370.981	13
73	0.3846	0.9712	367.715	13
74	0.4079	1.0230	368.747	13
75	0.3402	0.9812	369.504	13
76	0.4990	1.0150	365.744	13
77	0.3876	1.0018	368.674	13
78	0.3242	0.9725	369.742	13
79	0.3676	1.0088	369.553	13
80	0.3460	0.9720	368.997	13
81	0.3142	1.0131	371.475	13
82	0.3458	1.0405	371.329	13
83	0.4719	0.9571	364.560	13
84	0.3476	1.0169	370.480	13
85	0.3428	0.9632	368.796	13
86	0.4871	1.0071	365.823	13
87	0.3945	1.0285	369.345	13
88	0.4615	0.9690	365.280	13
89	0.3194	0.9931	370.615	13
90	0.3847	0.9967	368.594	13
91	0.4458	1.0173	367.392	13
92	0.4968	0.9598	363.931	13
93	0.3805	0.9839	368.289	13
94	0.4723	0.9749	365.164	13
95	0.3380	0.9949	370.047	13

96	0.3844	0.9779	367.953	13
97	0.3500	1.0423	371.249	13
98	0.3886	1.0361	369.791	13
99	0.4101	0.9551	366.354	13
100	0.4999	1.0336	366.330	13

Trial	x_1	P (atm)	T (K)	Iter.
101	0.4938	1.0426	366.800	13
102	0.4697	0.9666	364.957	13
103	0.3971	0.9714	367.331	13
104	0.3802	0.9559	367.325	13
105	0.3758	1.0485	370.609	13
106	0.3530	1.0284	370.682	13
107	0.3910	0.9923	368.240	13
108	0.4915	1.0495	367.093	13
109	0.4112	1.0218	368.607	13
110	0.3310	0.9797	369.766	13
111	0.4937	1.0079	365.659	13
112	0.4084	1.0248	368.790	13
113	0.3114	1.0084	371.408	13
114	0.4006	1.0353	369.382	13
115	0.3315	1.0461	371.988	13
116	0.3160	0.9686	369.882	13
117	0.4190	1.0175	368.216	13
118	0.3470	0.9620	368.613	13
119	0.4781	0.9746	364.987	13
120	0.4189	1.0119	368.033	13
121	0.3838	1.0084	369.016	13
122	0.4046	1.0435	369.522	13
123	0.3409	1.0216	370.865	13
124	0.3477	0.9896	369.547	13
125	0.4343	0.9800	366.476	13
126	0.3632	1.0252	370.242	13
127	0.3145	0.9958	370.877	13
128	0.4997	1.0496	366.855	13
129	0.3147	0.9713	370.023	13
130	0.3530	1.0433	371.176	13
131	0.4762	1.0379	367.166	13
132	0.3739	0.9658	367.874	13
133	0.4667	1.0204	366.861	13
134	0.4223	1.0487	369.144	13
135	0.4308	0.9508	365.567	13
136	0.4634	0.9799	365.598	13
137	0.4327	1.0439	368.668	13
138	0.3269	0.9615	369.266	13
139	0.3214	1.0053	370.963	13
140	0.3545	1.0105	370.035	13
141	0.4435	0.9704	365.866	13
142	0.4268	0.9764	366.580	13
143	0.3977	1.0405	369.644	13
144	0.4692	0.9592	364.713	13
145	0.3847	0.9777	367.941	13

146	0.3007	1.0271	372.409	13
147	0.4274	0.9762	366.556	13
148	0.4482	1.0052	366.910	13
149	0.3855	0.9510	366.983	13
150	0.3150	1.0383	372.287	13

Trial	x_1	P (atm)	T (K)	Iter.
151	0.4808	1.0046	365.921	13
152	0.4669	1.0083	366.458	13
153	0.3296	0.9627	369.217	13
154	0.3617	1.0399	370.786	13
155	0.4592	1.0361	367.606	13
156	0.4798	0.9710	364.810	13
157	0.3499	0.9603	368.460	13
158	0.4560	1.0384	367.776	13
159	0.3813	1.0121	369.223	13
160	0.3309	1.0430	371.902	13
161	0.4729	1.0476	367.575	13
162	0.4622	1.0381	367.587	13
163	0.3050	1.0237	372.147	13
164	0.3664	1.0431	370.731	13
165	0.4604	1.0364	367.581	13
166	0.4621	0.9767	365.524	13
167	0.4575	0.9608	365.115	13
168	0.4744	1.0359	367.148	13
169	0.3445	1.0317	371.078	13
170	0.3921	0.9805	367.807	13
171	0.4591	0.9728	365.482	13
172	0.3047	0.9693	370.291	13
173	0.3657	1.0364	370.535	13
174	0.4934	0.9779	364.651	13
175	0.4283	0.9900	367.001	13
176	0.4962	1.0036	365.445	13
177	0.4878	0.9615	364.249	13
178	0.4941	0.9679	364.291	13
179	0.4925	0.9765	364.633	13
180	0.3217	0.9935	370.554	13
181	0.4457	0.9814	366.177	13
182	0.4212	1.0011	367.599	13
183	0.3770	1.0077	369.211	13
184	0.3509	1.0209	370.505	13
185	0.3003	1.0426	372.934	13
186	0.4077	1.0219	368.716	13
187	0.4484	1.0171	367.300	13
188	0.3728	0.9570	367.599	13
189	0.4328	0.9830	366.623	13
190	0.3628	1.0348	370.578	13
191	0.4440	0.9800	366.183	13
192	0.3619	0.9908	369.125	13
193	0.3805	0.9796	368.143	13
194	0.3255	0.9920	370.377	13
195	0.4881	1.0177	366.153	13

196	0.4806	1.0116	366.165	13
197	0.3602	1.0048	369.656	13
198	0.3001	0.9787	370.779	13
199	0.3860	1.0080	368.936	13
200	0.4309	0.9965	367.142	13